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EX:3 write a python program to solve Water jug problem

AIM

To write a Python program to solve the Water Jug Problem using the Breadth-First Search (BFS) algorithm**,** where the user enters the capacities of two jugs and the required target amount.

ALGORITHM

1. Start the program and import deque for BFS.
2. Read the capacities of Jug 1, Jug 2, and the target amount.
3. Initialize both jugs as empty with state (0, 0).
4. Add the initial state to the queue and create a visited set.
5. Remove a state from the queue and check whether it has already been visited.
6. Check whether either jug contains the target amount; if yes, display “Goal reached!”.
7. Generate all possible states by filling, emptying, and pouring water between the two jugs.
8. Add the new states to the queue and repeat until the target is reached.

PROGRAM CODE

```
from collections import deque

a = int(input("Enter capacity of Jug 1: "))

b = int(input("Enter capacity of Jug 2: "))

target = int(input("Enter target amount: "))

queue = deque([(0, 0)])

visited = set()

while queue:

    x, y = queue.popleft()

    if (x, y) in visited:
```

```
        continue
    visited.add((x, y))
    print(x, y)
    if x == target or y == target:
        print("Goal reached!")
        break
    states = [
        (a, y),
        (x, b),
        (0, y),
        (x, 0),
        (x - min(x, b-y), y + min(x, b-y)),
        (x + min(y, a-x), y - min(y, a-x))
    ]
    queue.extend(states)
```

OUTPUT

```
============================
Enter capacity of Jug 1: 4
Enter capacity of Jug 2: 9
Enter target amount: 5
0 0
4 0
0 9
4 9
0 4
4 5
Goal reached!
```

RESULT

Thus, the Water Jug Problem was successfully solved using the Breadth-First Search (BFS) algorithm, and the required target amount of water was obtained.